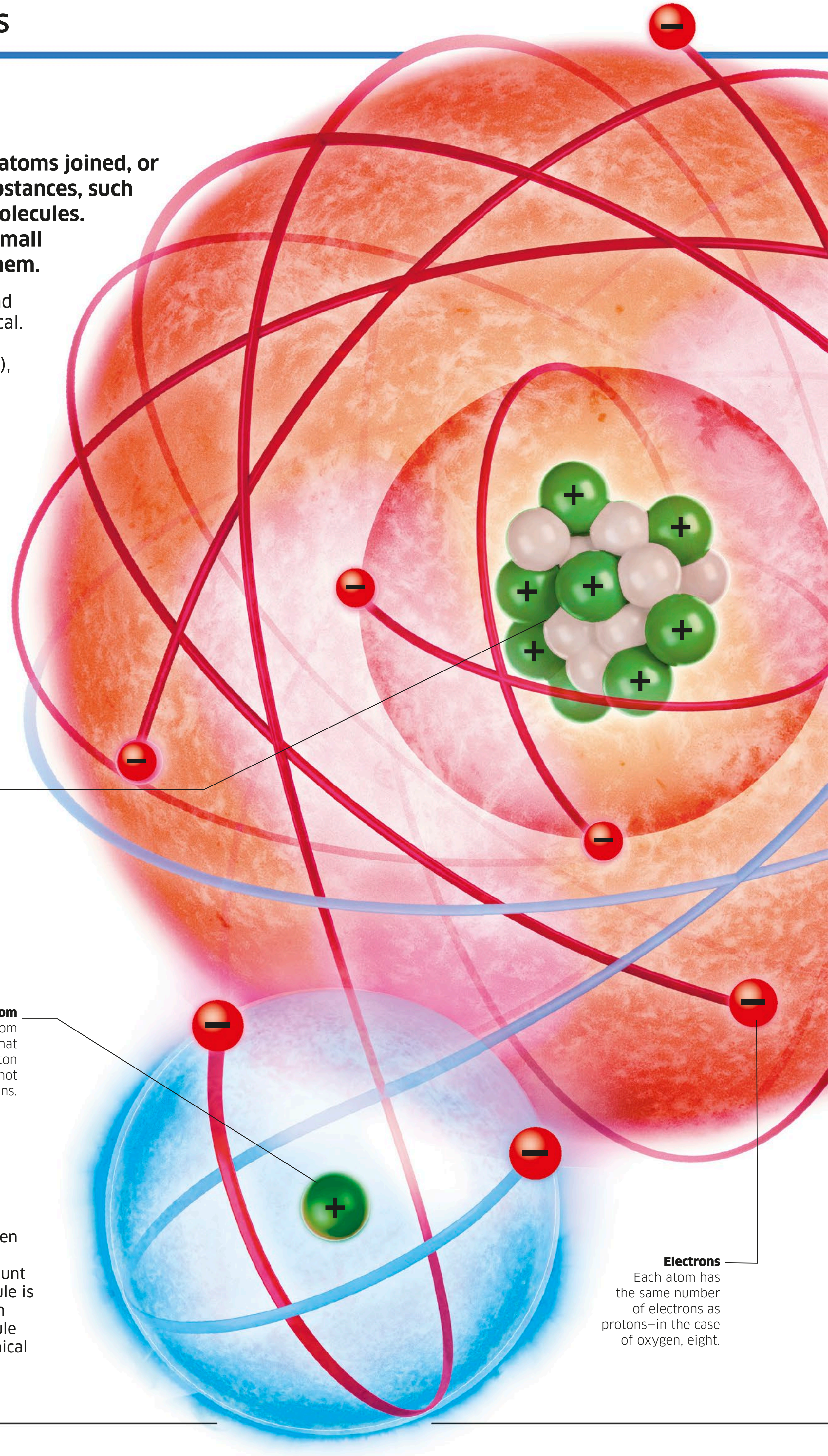
Molecules

A molecule consists of two or more atoms joined, or bonded, together. Many familiar substances, such as sugar or water, are made up of molecules. Molecules are so small that even a small drop of water contains trillions of them.

All the molecules of a particular compound (chemically bonded substance) are identical. Each one has the same number of atoms, from at least two elements (see pp.28–29), combined in the same way. The bonds that hold molecules together form during chemical reactions but they can be broken as atoms react with other atoms and rearrange to form new molecules. It is not only compounds that can exist as molecules. Many elements exist as molecules, too, but all the atoms that make up these molecules are identical, such as the pair of oxygen atoms that make up pure oxygen (O_2).



Water molecule

Imagine dividing a drop of water in half, and then in half again. If you could keep doing this, you would eventually end up with the smallest amount of water: a water molecule. Every water molecule is made up of one oxygen atom and two hydrogen atoms. The atoms are held together as a molecule because they share electrons, in a type of chemical bond called a covalent bond (see also p.16).